Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS **5054/21**

Paper 2 Theory **October/November 2016**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

© UCLES 2016

Section A

1 **(a)** velocity / it has a direction / is a vector — B1

(b) **(i)** (*F* =)*ma* **or** 800 × 1.5 — C1
1200 N — A1

(ii) friction / air resistance acts on car — B1
opposes force due to engine — B1

(iii) (Δ*v* =)*at* **or** 1.5 × 4.0 **or** 6.0 — C1
31 m / s — A1 [7]

2 **(a)** 260 N — B1

(b) **(i)** for a body in equilibrium — B1
(total) clockwise moment = (total) anticlockwise moment — B1

(ii) $F_1d_1 = F_2d_2$ **or** 260 × 0.35 **or** 91 **or** $F \times 0.65$ — C1
260 × 0.35 = $F \times 0.65$ **or** 260 × 0.35/0.65 **or** 91 = $F \times 0.65$ **or** 91/0.65 — C1
140 N — A1 [6]

3 **(a)** chemical (potential energy) — B1

(b) **(i)** non-renewable **and** oil / it is not replaced / will run out — B1

(ii) acid rain **or** produces CO_2 **or** warms lakes / rivers / sea **or** global warming **or** greenhouse effect — B1

(c) **(i)** <u>useful</u> energy output / (total) energy input **or** power for energy twice — B1

(ii) **1** $1.9 \times 10^9/0.38$ **or** $1.9 \times 10^9 \times 100/38$ — C1
5.0×10^9 W — A1
2 (*E* =)*Pt* **or** $0.62 \times 5.0 \times 10^9 \times 2.0$ (× 3600) **or** $(5.0 - 1.9) \times 10^9$ etc. — C1
2.2×10^{13} J — A1 [8]

4 **(a)** smallest angle for total internal reflection **or** angle for refraction along surface — B1
angle of <u>incidence</u> in (optically) <u>denser</u> medium — B1

(b) vertical ray continues undeviated — B1
second ray (60° to horizontal) refracts away from normal into the air — B1
third ray reflects internally **and** *i* = *r* by eye **not** if any refracted ray — B1 [5]

© UCLES 2016

5	**(a)**		number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b)	**(i)**	(λ =)c / f **or** 330 / 2200	C1	
			0.15 m	A1	
		(ii)	**1** no change **and**		
			2 increases	B1	
	(c)	**(i)**	**1** loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules)	B1	
			2 compressions **and** rarefactions / molecules vibrate / longitudinal wave	B1	
			vibration / oscillation / energy passed on	B1	
		(ii)	fewer / no molecules / particles **and** less / no energy / vibration transferred	B1	[8]
6	**(a)**	**(i)**	X N-pole	B1	
			Y S-pole **and** Z N-pole	B1	
		(ii)	they touch / move towards each other **and** opposite poles attract	B1	
	(b)		any sensible use: starting-motor circuit; with a logic gate; nuclear power station	B1	
			corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1	[5]
7	**(a)**	**(i)**	supplies the (mains) e.m.f. / voltage	B1	
		(ii)	to complete the circuit / is at 0 V	B1	
	(b)	**(i)**	the circuit / supply is cut / broken **or** current stops	B1	
			fuse melts / blows / burns	B1	
		(ii)	live wire	B1	
			when it cuts the circuit / melts no part of the appliance is live / no shock	B1	[6]

[45]

© UCLES 2016

Section B

8 (a) (i) 11 protons **and** 11 electrons B1
13 neutrons B1
electrons in orbit / surrounding nucleus **or** neutrons **and** protons in nucleus B1

(ii) **one** more neutron (in sodium-24) **or** one fewer neutron in sodium-23 B1 [4]

(b) (i) electron B1

(ii) $^{0}_{-1}(\beta)$ **cao** B1
^{24}(Mg) B1
$_{12}$(Mg) B1 [4]

(c) electromagnetic (radiation / rays / waves) M1
(very) high frequency / energy **or** (very) short wavelength A1 [2]

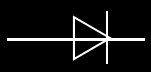
(d) (i) path curving upwards B1

(ii) path continues in straight line B1

(iii) beta-particle charged **or** gamma-ray uncharged B1 [3]

(e) long enough to take measurements **or** short enough so the body is not irradiated for long B1 B1 [2]
[15]

9 (a) (i) magnetic field mentioned B1
alternating / changing magnetic field B1
current / voltage / e.m.f. induced (in secondary coil) B1

(ii)  B1
diode B1 [5]

(b) (i) work done / energy transferred per unit charge M1
electrical energy to other forms **or** for whole circuit **or** property of supply A1

© UCLES 2016

(ii) **1** 1.3 V B1

2 (I =)V/R **or** 1.3/5.2 C1

0.25 A A1

3 (Q =)It **or** 0.25 × 1.5 × 3600 **or** 0.25 × 1.5 C1

0.25 × 1.5 × 3600 **or** 0.37/0.375/0.38 C1

1300 / 1350 / 1400 C A1 [8]

(c) plastic / casing is an (electrical) insulator M1

no shock possible A1 [2]

[15]

10 **(a)** **(i)** molecules / they close together **or** small gaps between molecules B1

(ii) molecules / they exert large (repulsive) forces B1 [2]

(b) **(i)** (V =)m/ρ **or** 680 / 0.85 C1

800 cm^3 **or** 8.0×10^{-4} m^3 A1

(ii) **1** molecules <u>vibrate</u> molecules <u>vibrate</u> B1

collide with neighbours **or** collide with electrons B1

transfer vibration / energy electrons travel through metal B1

2 heated / hot oil expands / less dense B1

rises B1

convection current/circulation established B1

3 any suitable named insulator **and** it is a poor conductor B1 [9]

(c) **(i)** <u>temperature</u> at which (liquid) vaporises / becomes gas / steam B1

(ii) (Q =)$mc\,\Delta T$ **or** 680 × 2.0 × (260 – 20) **or** 680 × 2.0 × 240 C1

3.3×10^5 J A1

(iii) heat supplied to <u>pan</u> **or** heat lost to <u>air / surroundings</u> B1 [4]

[15]

© UCLES 2016